

Salmon Stream Network Analysis

Introduction

This lab explores the Hydrology, Network Analyst, and Linear Referencing toolsets in ArcGIS Pro. A network is a representation of a process via a collection of nodes which are connected by edges. Both the nodes and edges have attributes that determines the behavior of a network. A hydrological basin is an example of a system which can be mapped using network analysis (Sarker et al., 2019). Nodes can be the intersections of streams and rivers, as well as other reservoirs such as lakes or ponds, or even obstructions such as dams. Edges represent flow dynamics, with features including channel depth, slope gradient, and water velocity impacting how water moves from node to node. Hydrological characteristics are important for a number of biophysical ecosystem processes, including the spawning of salmon, which require particular substrate materials, water quality, and minimum volatility in the water column in order to travel and lay their young. Linear referencing is a method of route determination along networks which ascribes an attribute along a route as a product of distance along the route (Pickell, 2026). The farther the stream is from the ocean, for example, the more difficult it is for salmon to reach it. In this way this lab will evaluate preferred habitat for salmon spawning on Vancouver Island.

Methods

This lab utilizes a depressionless digital elevation model (DEM) of Vancouver Island for hydrological modelling. Common practice in hydrological modeling, the removal of sinks to create a depressionless DEM removes ‘internal drainage’, that is low points where water can accumulate but are not part of the larger hydrological network as they do not connect anywhere (School of Forest Resources, 2010). This DEM is clipped to a boundary of Vancouver Island before producing Flow Direction and Flow Accumulation rasters using the tools by the same name in ArcGIS Pro. Next the Con tool is utilized to generate a binary raster so that a pixel with a Flow Accumulation count ≥ 1000 is assigned a value of 1. This output is used to generate Stream Links and Stream Order, using the tools of that name. Next, the Stream to Feature tool will convert the raster output to polylines. Finally, convert the stream features into routes using Create Routes.

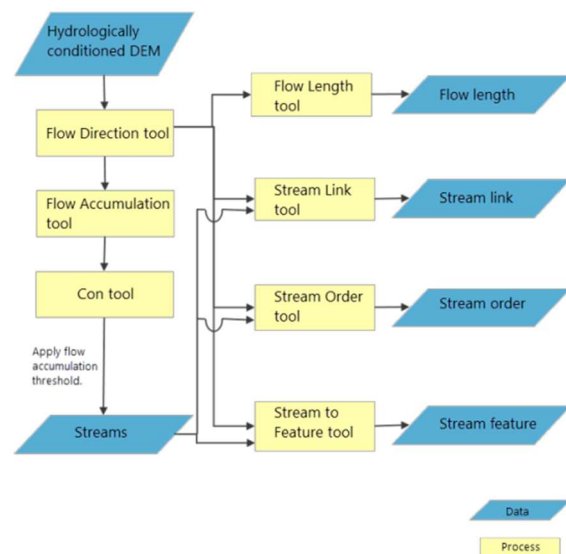


Figure 1. Workflow to produce a stream network in ArcGIS Pro.

The dams dataset represents dams as lines across the waterbodies. Use the Create Topology Wizard to split the streams where they intersect dams. Export the point errors to mark where the dams are along the stream network. Finally, use a negative buffer (nibble) to save a polyline of the Vancouver Island boundary 25 meters inland. Using the same topology process, export the points where the stream network intersects the nibble, representing where streams flow to the ocean.

Linear referencing is the process of assigning attributes to routes. Rasters of slope gradient and stream order will be extracted using Stream to Feature and overlaid onto the network using the Locate Features Along Routes tool. Finally, we want to trace the stream network we created to understand where salmon can go from the ocean inland. The result is a binary classification of the stream network as reached or unreachable. I chose to investigate the Juan de Fuca-Pachena conservation unit for coho salmon habitat, and present my findings below.

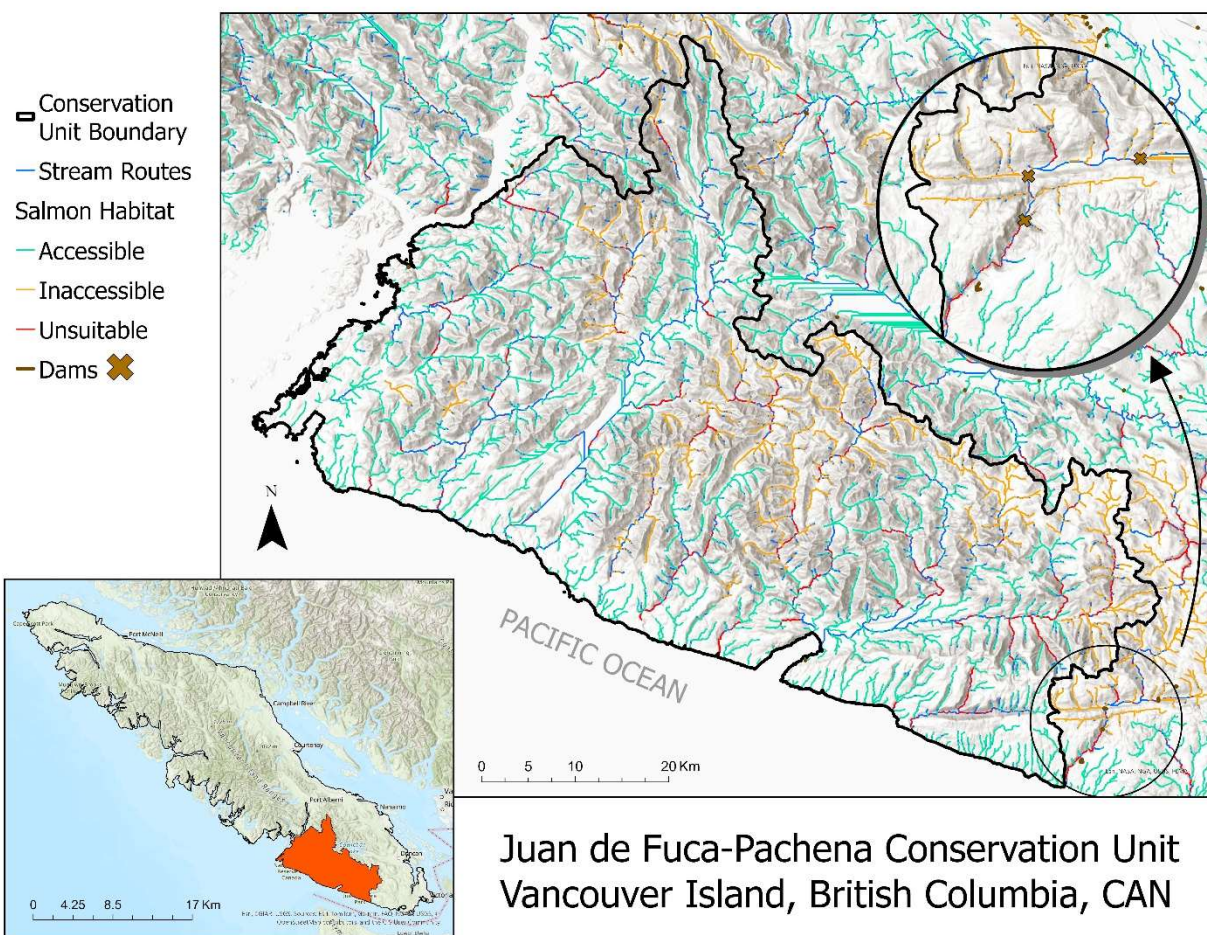


Figure 1. Map of the stream routes in the Juan de Fuca-Pachena Conservation Unit. Suitable salmon habitat was defined as streams with a slope incline less than or equal to 20% and that were stream order 1 or 2. These streams were divided into 'Accessible' streams if they met the habitat criteria and were reached by the Tracing algorithm, and 'Inaccessible' if they met the habitat criteria but were not reached. Stream segments that were both too steep and not of stream order 1 or 2 were denoted 'Unsuitable,' and the remaining streams that neither met nor failed both criteria were retained as generic 'Stream Routes.'

Results

The Juan de Fuca-Pachena Conservation Unit is located on the Southwest of Vancouver Island. The conservation unit, bordered by the Pacific Ocean to the South and an inlet to the West has an abundance of stream-ocean junctions, 222 specifically, across the Southern and Western edges of the unit.

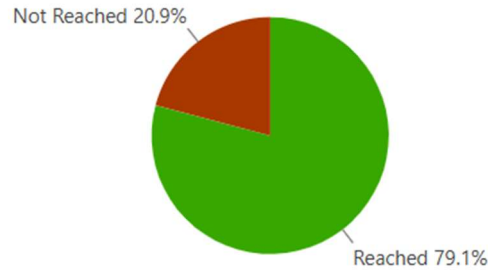


Figure 2.
Percentage of Reachable Stream by Total Length

Access and Connectivity

Salmon have relatively high freedom of movement and access to suitable habitat in the Juan de Fuca Conservation Unit. By length, 79.1% of streams in the study area are reachable by salmon (see Figure 2). Investigating Figure 1 reveals that many of the streams of suitable habitat for salmon spawning that are inaccessible are so because of their disconnection with the stream network that connects to the ocean. This represents perhaps errors in the generation of the stream network. In other places, it does seem accurate that Unsuitable streams, those with a gradient greater than 20% and a stream order larger than 2, as potentially untraversable means that habitat areas beyond are unreachable. Additionally, while the study area does not have any dams, just next door a handful of dams not actually in the study area restrict the ability for salmon to reach suitable habitat as there is no corridor to reach the streams (see Figure 1).

Habitat

Using gradient and stream order to define salmon habitat, there is a total of 15,221,018.3 m of salmon habitat in the study area, which amounts to 55.7 % of total stream length. Of this habitat, 11,846,595.3 m (77.83%) is accessible, and 3,374,423 m (22.17%) is not. By investigating Figure 1 we can also conclude that the accessible habitat is really clustered towards the ocean.

Table 1.

Percent accessibility of stream categories, where suitable salmon habitat is 55.7% of total streams in the conservation unit, by length.

	Accessible	Inaccessible
All Streams	79.1%	20.9%
Habitat (55.7% of total)	77.83%	22.17%

Lakes and Waterbodies

Directly to the north of the study area boundary lies Lake Cowichan, where many straight lines of accessible suitable habitat run parallel to each other. This is the case for all lakes and bodies of water across the stream network. Because we developed the stream network from flow direction and accumulation metrics, the stream network flow into and across all water bodies, but does so in individual strips. Instead of this we could, say, use a polygonal waterbody layer and join it to the stream network.

Discussion

The results of this lab reveal some of the challenges involved in hydrological modelling. Slight variances in workflow can produce different networks, and the model is only as robust as its inputs.

Creating a Network

As identified in the Results section, many of the streams were inaccessible in the study area. Upon closer examination, some of these are caused by omission or topological errors when generating the stream network, visible in Figure 3 below. While the streams for the most part follow existing ones, by product of the resolution of the DEM we used, the line denoting the stream is often shifted to one side or another, based on how the pixelated DEM generated flow direction and accumulation statistics. Alternatively, some streams are fully unconnected to the ocean-reaching stream network within the study area. When creating statistics, this skews the results as some of the streams are not able to be connected, and should be omitted.

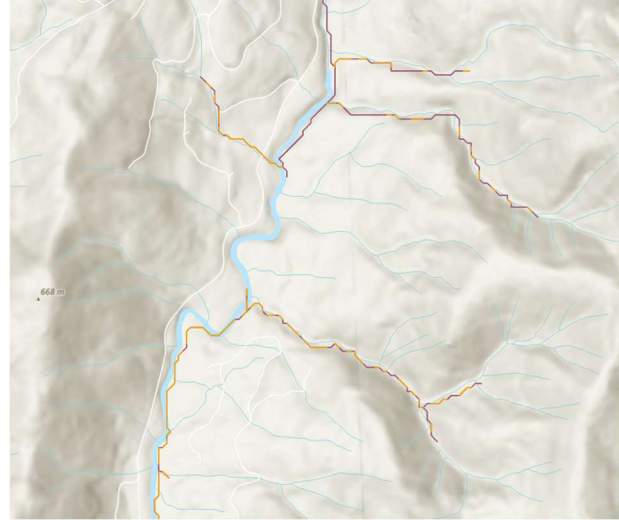


Figure 3. Example of a missed stream route just north of Edinburgh Mountain. The large river is fully disconnected in the middle. Additionally, many small streams (in blue) are not within the stream network (orange and red).

Increasing the resolution of an input DEM will heighten the ability to accurately identify the precise location of stream channels as the smaller cells will capture more starkly the changes in elevation over space. This will allow the small cell size of the hydrological tool outputs to be more accurate. Increasing the size of the raster, on the other hand, will decrease the detail that is able to be grasped, and smaller streams in particular will suffer. It might be fine if the Fraser River, or something on that scale is the object of interest, but in this study area a lot of the small streams are important habitat area, and so we would not want to obscure these opportunities for salmon spawning by manipulating the spatial resolution too inappropriately.

Access and Connectivity for Salmon Populations

My results more than anything demonstrate the fragility of stream networks to barriers. Even by product of topological errors while generating my stream network, streams experience extreme fracturing which shatters access and mobility to very vulnerable processes, such as salmon spawning. Salmon are born in freshwater, live in saltwater, and return to freshwater streams to lay their young. Amid persisting risks to wild salmon by an over-harvesting fishing industry, the key to the regeneration of salmon populations is their safe passage to the streams where their young are born. A key component of this is amid predators, environmental hazards, and other human activities, salmon must both make it to the streams to lay their young, and the young salmon must journey out to the ocean again. Reach – or the time and distance from a point to another is as essential measure of what habitat salmon can access from the ocean.

This lab used reach analysis to determine the streams available for salmon spawning using solely gradient and stream order as habitat criteria, and dams as barriers. In reality there are many more elements that restrict or enable salmon spawning in a stream. Salmon prefer coarse gravel, for example, to spawn their eggs in as they can move the particles (DFO, 2020). The presence of riparian vegetation, or vegetation along the banks of a stream heightens the likelihood of coarse gravel substrate, as the vegetation prevents erosion of smaller sediments into the stream. Water flow, also, is important as it cannot be so fast that salmon cannot swim, but it is important for there to be solid waterflow to flush waste. 1.0m/sec has been identified by the DFO as the maximum flow discharge that allows salmon to swim upstream (Levy & Slanley, 1993). In practice, coho salmon can withstand a maximum velocity of 2.44 m/sec, but require a minimum depth of 0.18m to swim in a given stream.

Future Studies

In the future, studies can incorporate additional salmon habitat characteristics, such as the ones listed by diversifying data sources. Water velocity and substrate type can be identified by a field station. Drone footage has also been used to quantify water velocity through a series of surface flow velocity equations (Saletti, 2024). Riparian vegetation can be identified through aerial imagery or satellite-based multispectral imagery, and finally stream depth can be either recorded manually, with bathymetric LiDAR, or RADAR.

Conserving viable salmon habitat on Vancouver Island is very important as the tree-like nature of the streams means that one barrier can restrict significant amounts of habitat. In the Juan de Fuca-Pachena conservation area, I would suggest monitoring for extremes in droughts or flooding that might render streams unsuitable for salmon. Legislation requiring clearance for bridge-building or that a fish ladder be constructed for all obstructions will ensure that salmon maintain access to their key habitat areas.

References

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